FEDERAL SPECIFICATION

DISINFECTANT-DETERGENT, GENERAL PURPOSE, CONCENTRATE (QUATERNARY AMMONIUM COMPOUND)

This specification was approved by the Commissioner, Federal Supply Service, General Services Administration, for the use of all Federal agencies.

1. SCOPE AND CLASSIFICATION

1.1 Scope. This specification covers free-flowing solid, and liquid form of quaternary ammonium compound disinfectant-detergent concentrate intended for general purpose building and hospital maintenance.

1.2 Classification. The disinfectant-detergent concentrate shall be of the following types and classes as specified (see 6.2).

Type I - Liquid.
Class 1 - Essential oil scented.
Class 2 - Pine oil scented.

Type II - Powder, premeasured water-soluble packet (essential oil scented).

2. APPLICABLE DOCUMENTS

2.1 The following documents, of the issues in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein.

Federal Specification:

L-P-391	-	Plastic Sheets, Rods and Tubing, Rigid, Cast, Methacrylate (Multiapplication).
O-S-642	-	Sodium Phosphate, Tribasic, Anhydrous; Dodecahydrate; and Monohydrate; Technical.
GG-H-80	-	Handle and Blade Set, Surgical Knife, Detachable Blade.
QQ-S-766	-	Steel Plates, Sheets, and Strip-Corrosion Resisting.
SS-T-312	-	Tile, Floor; Asphalt, Rubber, Vinyl, Vinyl-Asbestos.
TT-E-508	-	Enamel: Interior Semigloss, Tints and White.
TT-E-543	-	Enamel: Interior, Undercoat, Tints and White.
ZZ-G-421	-	Gloves, Surgeon.
LLL-F-1238	-	Floor Covering, Linoleum.
LLL-P-400	-	Pine Oil.
PPP-B-636	-	Boxes, Shipping, Fiberboard.
PPP-D-729	-	Drums, Shipping and Storage, Steel, 55-Gallon.
PPP-D-732	-	Drums, Metal, 55-Gallon Reconditioned (For Shipment of Noncorrosive Material).
PPP-P-704	-	Pails, Metal: (Shipping, Steel, 1 through 12 Gallon).
PPP-P-1655	-	Pail, Plastic, Shipping and Storage.

Federal Standards:

Fed. Std. No. 123 - Marking for Shipment (Civil Agencies).
Fed. Test Method Std. No. 536 - Soap and Soap-Products (Including Synthetic Detergents); Sampling and Testing.

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(Single copies of this specification and other Federal Specifications required by activities outside the Federal Government for bidding purposes are available without charge from Business Service Centers at the General Services Administration Regional Offices in Boston, New York, Washington, DC, Atlanta, Chicago, Kansas City, MO, Fort Worth, Denver, San Francisco, Los Angeles, and Seattle, WA.

(Federal Government activities may obtain copies of Federal Specifications, Standards, and Handbooks and the Index of Federal Specifications and Standards from established distribution points in their agencies.)

Military Standards:

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129 - Marking for Shipment and Storage.

(Copies of Military Specifications and Standards required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

Laws and Regulations:

16 CFR 1500 - Federal Hazardous Substances Act.
40 CFR 162 - Federal Insecticide, Fungicide, and Rodenticide Act.
49 CFR 171-179 - Department of Transportation Rules and Regulations for the Transportation of Explosives and Other Dangerous Articles.

(The Code of Federal Regulations (CFR) and the Federal Register (FR) are for sale on a subscription basis by the Superintendent of Documents, U.S. Government Printing Office, Washington, DC 20402. When indicated, reprints of certain regulations may be obtained from the Federal agency responsible for issuance thereof).

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless a specific issue is identified, the issue in effect on date of invitation for bids or request for proposal shall apply.

American Society for Testing and Materials (ASTM) Standard:

D 501 - Alkaline Detergents, Sampling and Chemical Analysis of.
D 523 - Specular Gloss.
D 1310 - Flash Point of Liquids by Tag Open-Cup Apparatus.
D 1429 - Specific Gravity of Water and Waste Water.
D 1767 - Ethylene Diamine Tetraacetate (EDTA) in Soap or Synthetic Detergents.
E 70 - pH of Aqueous Solutions with the Glass Electrode.
E 203 - Water Using Karl Fisher Reagent.

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA 19103.)

National Fire Protection Association (NFPA).

NFPA No. 56A

(Application for copies should be addressed to the National Fire Protection Association, 60 Battery-march Street, Boston, MA 02110.)

Association of Official Analytical Chemists (AOAC):

Official Methods of Analysis, Eleventh Edition, 1970.

(Application for copies should be addressed to the Association of Official Analytical Chemists, P.O. Box 540, Benjamin Franklin Station, Washington, DC 20044.)

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National Motor Freight Traffic Association, Inc., Agent:

National Motor Freight Classification.

(Application for copies should be addressed to the American Trucking Association, Inc., Tariff Order Section, 1616 P Street, N.W., Washington, DC 20036.)

Uniform Classification Committee, Agent:

Uniform Freight Classification.

(Application for copies should be addressed to the Uniform Classification Committee Room 1106, 222 South Riverside Plaza, Chicago, IL 60606.)

3. REQUIREMENTS

3.1 Materials.

TABLE I. Composition

Ingredients	Test Para.	Percent (minimum) Type I Class 1	Type I Class 2	Type II
Nonionic synthetic detergent, 100 percent active	4.4.14	3.0	3.0	10.0
Quaternary ammonium chloride (s)	4.4.16	3.0	3.0	11.4
Chelating agent, e.g. tetrasodium ethylenediamine tetraacetate	4.4.17	0.135	0.135	0.9
Essential oils	4.4.18	0.65	-	1.0
Pine oil	4.4.19	-	5.0	-
Alkaline builders, e.g. condensed phosphates	4.4.20	4.0	4.0	13.0
Water	4.4.21	-	-	None
Total nonvolatile solids	4.4.22	10.4	10.0	-
Isopropanol	-	-	5.0	-

3.1.1 Type I. Type I concentrate shall be a uniform liquid, free of sediments and suspended particles. The finished formulation shall meet the requirements as specified in section 3 and table I.

3.1.2 Type II. Type II concentrate shall be a clean, uniform, free flowing, granulated powder packaged in premeasured water-soluble packets made of poly-vinyl alcohol (PVA). Each 0.5 ounce packet concentrate shall contain 0.5 ± 0.01 ounce of powdered concentrate and each 1.5 ounce packet concentrate shall contain 1.5 ± 0.03 ounces of powdered concentrate. The finished formulation shall meet the requirements as specified in 4.4.9 and table I.

3.1.3 Excluded ingredients. The supplier shall furnish a certificate of compliance that the type I and type II concentrates contain no soap, sodium carboxymethylcellulose, formaldehyde or any of its polymeric forms, ammonia, anionic detergents, abrasives, bleaching agents, phenol or phenolics, iodine or iodophors, active halogens, mercury or heavy metals.

3.2 Test dilution. Except where otherwise indicated, the test dilution shall be the use-dilution specified by the supplier for the concentrate in distilled water which shall provide 450 ppm (minimum) of germicidal-fungicidal activity.

3.3 EPA Registration. The concentrates shall be registered with the Environmental Protection Agency (EPA) under the Federal Insecticide, Fungicide, and Rodenticide Act.

3.4 Biological requirements.

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3.4.1 Antimicrobial requirements (bactericidal and fungicidal action). Types I and II concentrates shall kill Staphylococus aureus (ATCC No. 6538), Salmonella typhosa (ATCC No. 6539), Tricophyton interdigitale (ATCC No. 640), and Pseudomonas aeruginosa (ATCC No. 15442) when tested as specified in 4.4.1.

3.4.2 Acute oral toxicity. The concentrate shall kill no more than five of a group of 10 rats (LD_{50} (oral) = 4 ml/kg. of body weight) when tested as specified in 4.4.2.

3.4.3 Dermatitis. Types I and II concentrates shall cause no dermatitis as defined in 16 CFR 1500.3 when tested in accordance with 16 CFR 1500.41 (see 4.4.3).

3.5 Deleterious effects.

3.5.1 Painted surfaces. Types I and II concentrates shall cause not greater than one-half the loss of 60 degree specular gloss of painted surfaces when tested as specified in 4.4.4.

3.5.2 Tile. Types I and II concentrates shall cause no softening, swelling, cracking, or discoloration of the test specimens when tested as specified in 4.4.5.

3.5.3 Plastics and rubber. Types I and II concentrates shall cause no stickiness, crazing, or change in color when tested as specified in 4.4.6.

3.5.4 Metal. Types I and II concentrates shall show no corrosive effect on stainless steel and not greater than 12 milligrams loss in weight of galvanized iron when tested as specified in 4.4.7 and 4.4.8, respectively.

3.5.5 Conductive flooring. Types I and II concentrates shall not adversely affect the conductivity of conductive flooring. The manufacturer shall submit evidence of Underwriters' Laboratory approval or furnish a certified copy of analysis performed by either an indepentent laboratory or by the manufacturers private facilities based on the electrical conductivity requirements of NFPA No. 56A.

3.6 Physical and chemical requirements.

3.6.1 Solubility. Types I and II concentrates shall be water-soluble in all proportions and shall form a clear, homogeneous solution when tested as specified in 4.4.10.

3.6.2 Flash point, type I. Type I, class 1 concentrate shall have no measurable flash point below 100° C. (212° F.) when tested as specified in 4.4.11. Type I, class 2 concentrate shall have no measurable flash point below 90° C. (194° F.).

3.6.3 pH. Types I and II concentrates shall have a pH 9.5 - 10.5 at the recommended use-dilution when tested as specified in 4.4.12.

3.6.4 Specific gravity, type I. The concentrate shall have a specific gravity of 1.03 - 1.08 when tested as specified in 4.4.13.

3.6.5 Stability in storage. Types I and II concentrates shall not lose original effectiveness and chemical stability, develop a rancid, fishy or acrid odor and show no tendency toward leakage when stored for 1 year in a closed shipping container maintained between temperatures of 40° F and 120° F. In addition, type I concentrate shall show no tendency toward precipitation and sedimentation.

3.6.6 Biodegradability. The synthetic detergent used in the manufacture of the product shall be 90 percent biodegradable, when tested as specified in 4.4.14.

3.6.7 Cleaning efficiency. Types I and II concentrates shall exhibit a comparative cleaning efficiency of not less than 60 percent when tested as specified in 4.4.15.

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3.6.8 Hard water effect. The germicidal activity of the concentrates shall not be affected by water of 500 ppm hardness when tested as specified in 4.4.23.

3.6.9 Workmanship. Types I and II concentrate shall be made from uniformly high grade raw materials in adequate proportions to insure conformity to the requirements of this specification. The concentrates shall be clean, uniform, free from foreign matter, and suitable for the intended use.

3.6.10 Labeling. The containers shall be marked in accordance with the Federal Insecticide, Fungicide, and Rodenticide Act and Department of Transportation Rules and Regulations and Federal Hazardous Substances Act.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure that supplies and services conform to prescribed requirements.

4.2 Certificates of compliance. Where certificates are submitted, the Government reserves the right to check-test such items to determine the validity of the certification.

4.3 Quality conformance inspections.

4.3.1 Examination of preparation for delivery. An examination shall be made to determine compliance with the packaging, packing, and marking requirements of section 5. Defects shall be scored as specified in table II. Sampling shall be in accordance with MIL-STD-105. The sample unit shall be one container fully prepared for delivery. The lot shall be the number of containers fully prepared for delivery. The lot shall be the number of containers offered for inspection at one time. The inspection level shall be S-2 with an AQL of 4.0 expressed in terms of defects per hundred units.

TABLE II. Examination of preparation for delivery

Examine	Defect
Containers	Not as specified.
Content	Not as specified.
Marking	Omitted; incorrect; illegible; improper size, location, sequence or method of application.
Material	Component missing or damaged.
Workmanship	Bulging or distortion of containers. Cushioning inadequate or improper.

4.3.2 Examination of closure, waterproofing and banding of containers. When shipping containers are required to comply with PPP-B-636 or PPP-P-704, examination for defects in closure, waterproofing or banding shall be in accordance with the appendix of that specification.

4.3.3 Examination of the end item. An inspection shall be made to determine conformance to the material and workmanship requirements of section 3. The sample unit for this inspection shall be the contents of one filled unit container as specified in 5.1.1.

Examine	Defect
Material	Type I, not a liquid. Type II, not a powder.
Workmanship	Not uniform; not free from foreign matter.

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4.3.4 Testing of the end item. Types I and II concentrates shall be tested for conformance to the requirements of section 3, in accordance with the test methods specified in table III. The lot size shall be expressed in units of packets for type I and in gallons for type II, of concentrate prepared from one batch, manufactured in a single production and offered for inspection at one time. The sample unit for type I shall be a one-gallon composite. The sample unit for type II shall be a one-ounce composite. The sample unit shall be thoroughly mixed to yield the composite sample and stored in a clean, dry, sealed, glass containers with identifying lot number and date. All tests shall be performed on the composite sample. Care shall be exercised during sampling, weighing, and storage to prevent loss or absorption of moisture or volatile material. Test report shall contain individual values utilized in expressing final result. The lot shall be unacceptable if one sample unit or the composite sample fail to meet any test requirement specified.

TABLE III. Directions for testing of the end item

Characteristic	Specification reference Requirement	Test Method	Number of Determinations	Results reported Pass/Fail	Results reported Numerically to Nearest
Antimicrobial requirements	3.4.1	4.4.1	1	X	
Acute oral toxicity,	3.4.2	4.4.2	1	X	
Dermatitis	3.4.3	4.4.3	1	X	
Effect on painted surfaces	3.5.1	4.4.4	1	X	
Effect on tile	3.5.2	4.4.5	1	X	
Effect on plastics and rubber	3.5.3	4.4.6	1	X	
Effect on stainless steel	3.5.4	4.4.7	1	X	
Effect on galvanized iron	3.5.4	4.4.8	1		0.001 gram
Conductive flooring	3.5.5	-	1		
Solubility	3.6.1	4.4.10	1	X	
Flash point, type I	3.6.2	4.4.11	1		Centigrade°
pH	3.6.3	4.4.12	1		0.10 pH-unit
Specific gravity, type I	3.6.4	4.4.13	1		0.0001 unit
Cleaning efficiency	3.6.7	4.4.15	2		0.1 percent
Hard water effect	3.6.8	4.4.23	1	X	

4.4 Test methods.

4.4.1 Antimicrobial requirements. Microbial activity shall be determined according to the AOAC method/paragraph specified below:

Microorganism	AOAC method/paragraph
Salmonella typhosa	Use-Dilution/4.001
Staphylococcus aureus	Use-Dilution/4.010
Tricophyton interdigitale	Use-Dilution/4.018
Pseudomonas aeruginosa	Use-Dilution/4.011

4.4.2 Acute oral toxicity. Ten healthy albino rats of the Wister strain, weighing between 150 to 250 gm shall be used for the oral toxicity test. The animals, none of whom shall be pregnant, shall be fasted for 24 hours preceding dosing. A wooden tongue depressor, narrowed at its central width and with a hole bored into its center large enough to admit a No. 10 French rubber catheter, shall be used as a gag to hold open the rat's mouth and permit entrance of the catheter. The test rats shall be weighted and a 1 ml tuberculin syringe to which a No. 10 French

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catheter has been attached shall be filled with a measured amount of the test dilution of the concentrate (4 ml. per kilogram of body weight). The rat shall then be held firmly with the other hand against the anterior teeth of the animal thus forcing the mouth open. Place the flat edge of the gauge against the tongue. By turning the gag through a half-turn (counter-clockwise), the tongue is pulled forward and the bored hole in the depressor is approximated with the pharynx. With the animal's jaws open, the depressor is locked behind the canine teeth. The forefinger of the operator's hand holding the rat is then hooked over the end of the gag. With the other hand, insert the catheter firmly through the hole in the gag, through the pharynx and esophagus and into the stomach. When certain that the tip of the catheter is in the stomach, depress the plunger of the syringe to deliver the calculated amount of concentrate. Withdraw catheter, remove gag, and place animal in normal environment with usual food and water. Keep the animals under observation for a period of 1 week.

4.4.3 Dermatitis. Patch test shall be determined in accordance with 16 CFR 1500.41.

4.4.4 Effect on painted surfaces. Test panels, approximately 3-1/2 x 6 x 1/16 inches (90 x 150 x 2 mm), conforming to QQ-S-766, shall be prepared by applying one coat of each of the following paints:

(a) Enamel undercoat, white, conforming to TT-E-543, as the first coat. A drying time of not less than 24 hours shall be allowed between the undercoat and the final coat.
(b) Enamel semi-gloss, white, conforming to TT-E-508, with a 60° specular gloss of 50 to 60, as the final coat. A drying time of not less than 48 hours and not more than 96 hours shall be allowed Specular gloss shall be measured in accordance with ASTM D 523.

Prepare one liter of a solution of trisodium phosphate, by dissolving 18 grams of the dodecahydrate, conforming to O-S-642, type II, in one liter of distilled water. Prepare one liter of the test dilution. Measure the specular gloss of each of the two panels, then immerse one panel in each of the above solutions, at room temperature, for one hour. At the end of this time, remove the panels, rinse thoroughly with tap water, give a final rinse with distilled water, and dry at room temperature at least one hour. Again measure the specular gloss of each panel. The percentage decrease in gloss of each panel shall be calculated by:

$$\text{Percent decrease in gloss} = \frac{G_1 - G_2 \times 100}{G_1}$$

where: G_1 = specular gloss before immersion
G_2 = specular gloss after immersion

4.4.5 Effect on tile. Test panels, approximately 2 x 4 x 1/8 inches (50 x 100 x 3 mm), as specified below, shall be used:

White linoleum, conforming to LLL-F-1238, type I, class 2, grade A.
Black asphalt, conforming to SS-T-312, type I.
Black rubber, conforming to SS-T-312, type II.
Black vinyl-asbestos, conforming to SS-T-312, type IV.

Place 150 ml of the test dilution into each of four 250 ml beakers. Immerse one panel in 1 beaker containing the test dilution. Allow the panels to remain in the dilution at room temperature for 6 hours. At the end of this time, remove the panels, rinse thoroughly with tap water, give a final rinse with distilled water, and dry at room temperature at least 1 hour. Examine for softening, swelling, cracking, or discoloration.

4.4.6 Effect on plastic and rubber. A 2-inch square of methacrylate plastic, conforming to L-P-391, and a rubber glove, conforming to ZZ-G-421, grade A, shall be completely immersed in the test dilution for four hours at room temperature (20 - 25° C). After that time, the items shall be removed, thoroughly rinsed with tap water, and given a final rinse in distilled water. Allow to air dry and examine for softening, swelling, cracking, or discoloration.

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4.4.7 Effect on stainless steel. A scalpel blade conforming to GG-H-80 shall be completely immersed in the test dilution for 16 hours at 25° C. At the end of this time the blade shall be removed, thoroughly rinsed with tap water and given a final rinse in distilled water. Dry and examine for signs of corrosion.

4.4.8 Effect on galvanized iron. Six 2 x 2 inch (50 x 50 mm) test panels shall be cut from 18 gauge galvanized iron sheet and washed in carbon tetrachloride. Prior to the test, the test panels shall be wiped with a lint-free clean cloth moistened with carbon tetrachloride, dried at 105° C for 30 minutes, and cooled in a dessicator. Each test panel shall be accurately weighed to the nearest 0.1 milligram and completely immersed, 2 to a beaker, in three beakers containing the test dilution. The test panels shall not touch each other while immersed. The beakers shall be covered with a watch glass and immersed to the solution level in a constant temperature water bath maintained at 35 + 1° C for 6 hours. At the end of this time, the panels shall be removed from the dilutions, rinsed thoroughly with tap water, given a final rinse with distilled water, wiped with a clean lint-free cloth, and dried at 105° C for 30 minutes. The panels shall be cooled in a dessicator and reweighed. The weight loss shall be calculated by:

$$\text{Average weight loss} = \sum_{i=1}^{6} \frac{(W_1 - W_2)}{6}_i$$

where: W_1 = weight before immersion.
W_2 = weight after immersion.

4.4.9 Packet Product Weight, type II. Three concentrate packets shall be weighed to the nearest 0.01 oz. The product shall be removed from the packet and the packet shall be reweighed. The difference in weight shall fall between the limits of 3.1.2.

4.4.10 Solubility.

4.4.10.1 The type I concentrate and the type II packet concentrate when shaken for ten minutes in 4,000 ml (four liters) of synthetic hard water (0.264 grams of $CaCl_2.2H_2O$ and 0.295 grams of $MgSO_4.7H_2O$ in one liter of distilled water) at room temperature shall show no precipitation, stratification or other evidence of insolubility.

4.4.11 Flash point, type I. The flash point of the undiluted concentrate shall be determined in accordance with ASTM D 1310.

4.4.12 pH. The pH of the use dilution shall be determined in accordance with ASTM E 70.

4.4.13 Specific gravity, type I. The concentrate shall be determined in accordance with ASTM **D 1429.**

4.4.14 Nonionic synthetic detergent. Until an acceptable method of determining the biodegradiability of nonionic detergent is agreed upon and is acceptable to the Government, testing shall be performed in accordance with a method currently being used by the detergent industry for the evaluation of these compounds.

4.4.15 Cleaning efficiency. The cleaning efficiency shall be determined in accordance with Fed. Test Method Std. No. 536, Method 6701, except that the test dilution in 3.2 shall be used.

4.4.16 Quaternary ammonium chloride(s).

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4.4.16.1 Apparatus.

a. Bottles, 200 ml narrow mouth glass stoppered.
b. Burettes, 25 ml capacity.
c. Graduated cylinders, 25 ml, 10 ml.
d. Pipettes, volumetric, 10 ml capacity.
e. Volumetric flask.

4.4.16.2 Reagents. Where applicable, analytical grade chemicals and distilled water shall be used throughout the tests.

a. Anionic solution - sodium lauryl sulfate. Weigh 2.5 g to the nearest mg and dilute to 1000 ml in a 1 liter volumetric flask with distilled water. From the molecular weight, 296, and the percent activity which has been determined, calculate the normality (.008N).
b. Chloroform.
c. Bromphenol blue indicator solution. Dissolve 0.10 g of bromphenol blue indicator in 50 ml of ethanol and 50 ml of water.
d. Salt buffer solution. Dissolve 100 g of sodium sulfate and 10 g of sodium carbonate in water and dilute to 1 liter.

4.4.16.3 Procedure. Accurately weigh the required weight of sample (to the nearest mg) into a 250-ml beaker.

$$\frac{50 \times \text{Normality of titrant} \times \text{Mol wt. of Cationic}}{\% \text{ Expected Activity}} = \text{Grams of sample required}$$

Add 150 ml of water. Stir to dissolve, warming on the steam bath if necessary. Quantitatively transfer for sample solution to a 500-ml volumetric flask and dilute to volume with water at room temperature. Mix well. With a 10-ml volumetric pipette, transfer an aliquot of the diluted material to the 200-ml titration bottle. Into a 50-ml graduated cylinder, add 25 ml of chloroform and 25 ml of salt buffer solution and 5 drops of bromphenol blue indicator. Add this mixture to the bottle containing the aliquot of the diluted sample. Titrate the contents of the bottle with the standard anionic solution. At first, add the titrant in one ml increments, shaking the bottle vigorously for 5 - 10 seconds after each addition. As the end-point is approached, indicated by the increasing ease of separation of the layers, add suitable smaller increments of titrant. The endpoint is taken as the point at which the definite purple color appears in the upper aqueous layer.

4.4.16.4 Calculation.

$$\% \text{ Activity} = \frac{\text{ml anionic solution} \times \text{normality} \times \text{mol wt.}}{\text{wt. of sample in aliquot}}$$

4.4.17 Tetrasodium ethylenediamine tetraacetate. The tetrasodium ethylenediamine tetraacetate content of the concentrates shall be determined in accordance with ASTM D1767.

4.4.18 Essential oils. The essential oils content of the concentrates shall be determined in accordance with AOAC paragraph 19.117.

4.4.19 Pine oil. The pine oil content in the concentrate shall be determined in accordance with the test method specified in LLL-P-400.

4.4.20 Alkaline builders. The alkaline builders shall include silicates, carbonates and condensed phosphates (e.g. tetrasodium pyrophosphate). The condensed phosphates content shall be determined in accordance with ASTM D 501.

4.4.21 Water, type II. The water content shall be determined in accordance with ASTM E 203.

4.4.22 Nonvolatile solids, type I. Weigh approximately 100.0 g of concentrate into a tared flat-bottom dish approximately 8 cm in diameter provided with a tightly fitted cover. Remove the cover and place sample over a steam bath, for a minimum of 2 hours, until the bulk of the volatile matter has evaporated and then place in an air convection oven maintained at 105° C ± 2° C for 4 hours. Cool in a dessicator, replace cover, and weigh residue. Calculate percent nonvolatile matter as follows:

$$\text{Percent of nonvolatile matter} = \frac{\text{weight of residue} \times 100}{\text{weight of sample}}$$

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4.4.23 Hard water effect. A 200 ppm of the disinfectant-concentrates shall kill 99.999% of the test cultures of Staphylococus aureus (ATCC No. 6538) in waters of 500 ppm hardness when tested by the Germicidal and Detergent Sanitizers Method as outlined in the Official Methods of the Analysis of the Association of Official Agricultural Chemists.

5. PREPARATION FOR DELIVERY

5.1 Packaging. Packaging shall be level A, B or C as specified (see 6.2).

5.1.1 Level A and B.

5.1.1.1 Type I. Type I liquid disinfectant detergent of one class only shall be packaged as specified in 5.1.1.1.1 or 5.1.1.1.2.

5.1.1.1.1 Five gallons. Five gallons of product shall be furnished in a pail conforming to PPP-P-704 or PPP-P-1655.

5.1.1.1.2 Fifty-five gallons. Fifty-five gallons of product shall be furnished in a drum conforming to PPP-D-729 or PPP-D-732.

5.1.1.2 Type II. Type II powder disinfectant detergent of one quantity packet only, shall be packaged as specified in 5.1.1.2.1.

5.1.1.2.1 Packets. Forty one and one-half ounce or 96 one-half ounce packets shall be packaged in a water resistant canister with a reclosable lid.

5.1.2 Level C. The disinfectant detergent shall be packaged to afford adequate protection against damage during shipment from the supply source to the first receiving agency.

5.2 Packing. Packing shall be level A, B, or C, as specified (see 6.2).

5.2.1 Level A.

5.2.1.1 Type I. Disinfectant detergent packaged as specified in 5.1.1.1 does not require additional packing.

5.2.1.2 Type II. Nine canisters of disinfectant detergent packaged as specified in 5.1.1.2 shall be packed in a box conforming to PPP-B-636, class weather resistant. The box shall be closed, waterproofed and banded in accordance with the appendix to the box specification.

5.2.2 Level B. Type I and II disinfectant detergent shall be packed as specified in 5.2.1 except that for type II, the box shall be class domestic and the requirements for waterproofing and banding do not apply.

5.2.3 Level C. The disinfectant detergent in quantities as specified (see 6.2), packaged as specified in 5.1, shall be packed in containers to assure carrier acceptance and safe arrival at destination. Containers shall comply with the Uniform Freight Classification or National Motor Freight Classification Rules, as applicable.

5.3 Labeling and marking. The labeling and marking shall comply with the Federal Insecticide, Fungicide and Rodenticide Act and Regulations Promulgated Thereunder 40 CFR 162. Marking of the interior package and exterior shipping containers shall be in accordance with Fed. Std. No. 123 for civil agencies and MIL-STD-129 for military agencies, as applicable.

6. NOTES

6.1 Intended use. The concentrate covered by this specification is intended for use as a general disinfectant-detergent of environmental surfaces such as rubber, vinyl, vinyl asbestos, asphalt, linoleum, terrazzo, porcelain enamel, enamel painted or other surfaces which would be unaffected by contact with water. It is not intended for surgical use.

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6.2 Ordering data. Purchasers should select the preferred options permitted herein, and include the following information in procurement documents:

(a) Title, number, and date of this specification.
(b) Types and classes required (see 1.2).
(c) Quantities required (see 5.2.3).
(d) Selection of level of packaging and packing (see 5.1 and 5.2).
(e) Special marking required (see 5.3).

6.3 Definitions.

6.3.1 Batch. A batch is defined as that quantity of material which has been manufactured by some unit chemical process and subjected to physical mixing intended to make the final product substantially homogeneous.

6.3.2 Lot. A lot is defined as material produced by one manufacturer at one plant from the same raw materials and under essentially the same conditions by continuous or batch process. In the event the process is a batch operation, each batch shall constitute a lot.

MILITARY INTEREST:

Custodians:

Army - GL
Navy - SH
Air Force - 68

Civil Agencies Coordinating Activities:

GSA-FSS
HEW-NIH
VA - DMS
CPSC
USDA - GSD

Review Activities:

Army - MD
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PREPARING ACTIVITY: GSA-FSS

User activities:

Army - ME
Navy - YD, MC

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